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General Knowledge
World Geography

STUDY NOTES

Biogeography: Soil

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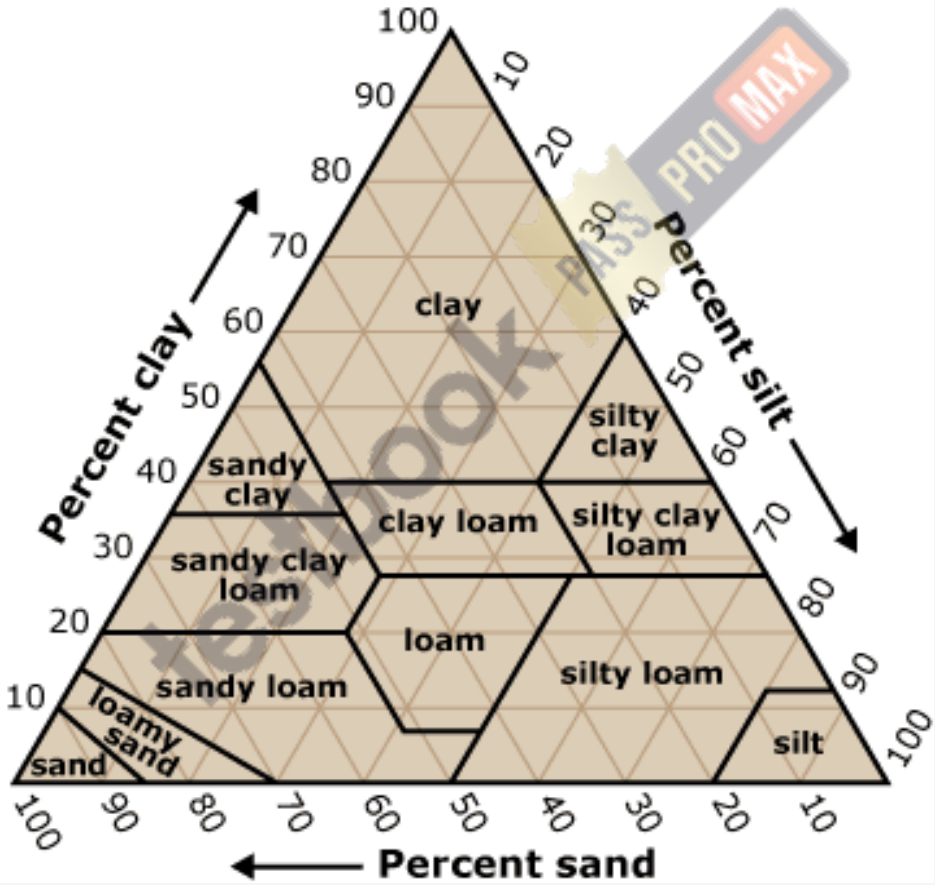
Soil

Soil is a mixture of minerals, organic matter, organisms, gases, and water that forms the thin top layer of the earth. It is biologically active, porous,s and a principal substratum for supporting life on Earth.

Pedology is the study of the formation, morphology, and classification of soils in their natural environment.

Soil Types

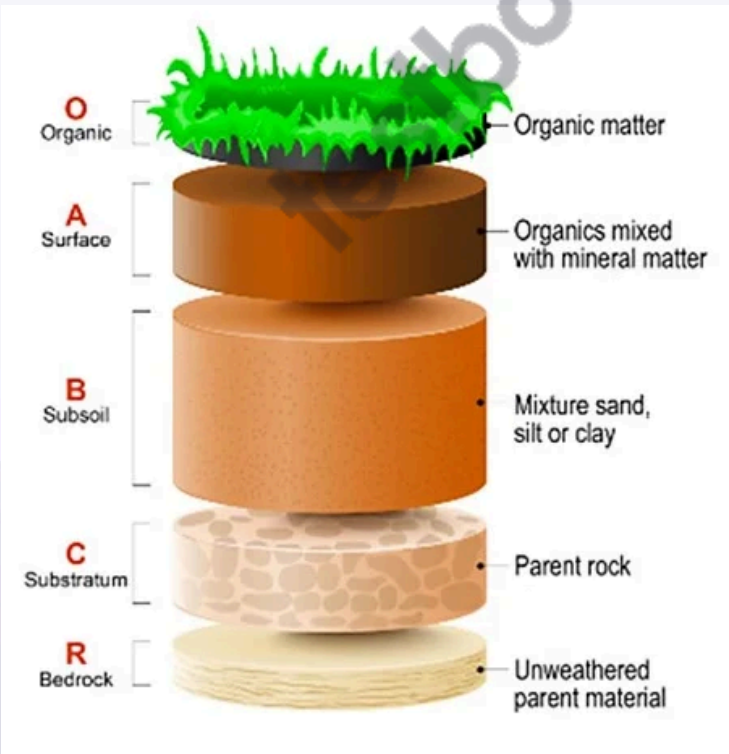


- **The soil is classified on the basis of the proportion of particles of various sizes.**
- If the soil contains a greater proportion of big particles, it is called **sandy soil.**
- If the proportion of fine particles is relatively higher, then it is called **clayey soil.**
- **Silt** occurs as a deposit in river beds. The size of the silt particles is between those of sand and clay. If the amount of large and fine particles is about the same, then the soil is called **loamy.**

- Water can drain quickly through the spaces between the sand particles. So, sandy soils tend to be light in color, well-aerated and dry.
- Clay particles, being much smaller, pack tightly together, **leaving little space for air**. Unlike sandy soil, water can be held in the tiny gaps between the particles of clay. So, clay soils have little air. But they are heavy as they **hold more water** than the sandy soils.
- The best topsoil for growing plants is loam. Loamy soil is a mixture of sand, clay and another type of soil particle known as **silt**. The loamy soil also has **humus** in it. It has the right water-holding capacity for the growth of plants.
- Depending upon the crop clay and loam soils are best suited for agriculture.
- Example –
  1. Clayey and loamy soils are both suitable for growing cereals like **wheat**, and **gram**. Such soils are good at retaining water.
  2. **Paddy** requires soils rich in clay and organic matter and having a good capacity to retain water is ideal.
  3. **Lentils** and other pulses, loamy soils, which drain water easily, are required.
  4. **Cotton**, sandy loam, or loam, which drains water easily and can hold plenty of air, are more suitable.

Soil Profile

- A vertical section through different layers of the soil is called the soil profile. Each layer differs in texture, color, depth and chemical composition.
- These layers are referred to as **horizons**. A soil horizon is a layer generally parallel to the soil surface, whose physical characteristics differ from the layers above and beneath.
- Horizons are defined in most cases by obvious physical features, chiefly color and texture.



Factors that influence soil formation

Parent Material

- The rocks from which soils are formed are called **parent materials**. In most of cases, the parent material determines the **coloration, mineral composition, and texture of the soil.**
- In some cases, the soil formed may or may not have the same physical properties as the parent rock as climatic factors induce chemical changes which also affect the physical properties of the soil.
- The surface rocks are exposed to the process of weathering. In this process, the rocks are converted into fine grains and provide a base for soil formation.

Relief

- The relief is one the most important factors for soil formation in places like mountain slopes and plateaus as can be seen on barren slopes where erosion is rampant, and it hinders soil formation. Example: **Chambal ravines**, higher reaches of the Himalayas where there is minimal or no forest cover.
- The areas of low relief or gentle slope generally experience deposition and have deep soils. Example: Indo-Gangetic plain.

Climate

- **Temperature and rainfall are the most important factors in soil formation.** They determine the effectiveness of weathering of the parent material, the quantity of water seeping through the soil, and the type of micro-organisms present therein.
- Two different parent materials may develop the same soil in the same type of climate. At the same time, the same parent material may produce two different types of soils in two different types of climates.
- **Hot summer and low rainfall develop black soil as is found in some parts of Tamil Nadu irrespective of the parent rock.** In Rajasthan, both granite and sandstone give birth to sandy soil under an arid climate.
- In arid and semi-arid regions, evaporation always exceeds precipitation. There is little vegetation, and the soils badly lack humus content. Hence the soils are invariably of **light colour.**

- ***Note - In areas of heavy rainfall and high temperature, the soils are red or lateritic. Why?***

- Rainfall during the **rainy season** washes the upper soil and leeches the materials into the deeper but during the dry summer season, the evaporation exceeds precipitation, and through capillary action, iron and aluminum oxides are transported to the surface making the soil **red**.
- In areas of alternate wet and dry climates, the leached material that goes deep down in the horizon is brought up, and the blazing sun bakes the topsoil so hard that it resembles a brick. Therefore, this soil is called **lateritic** which means brick.

Natural Vegetation

- Natural vegetation describes the combined effects of relief and climate. The formation and development of soil is very much influenced by the growth of vegetation.
- The decayed leaf material adds much-needed humus to soil thereby increasing its fertility.
- The densely forested areas contain some of the best soils in India. There is a close relationship between the vegetation types and soil types in India.